TRACE: Regret-Aware Instance Selection for Embodied Instruction Following

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***Abstract*—Embodied instruction following requires selecting a physical object instance from a class-level instruction. Identity confidence alone cannot determine whether to override a modular host's default selection, because the cost of a wrong mask depends on the scene. We present TRACE, a drop-in module for interaction-mask selection. TRACE derives task relations from the history of successful actions and obtains candidate beliefs from a decision model. A symbolic critic estimates candidate-specific extra interaction and navigation costs, which TRACE weights by the beliefs to compute expected regret. It overrides the host only when a unique candidate minimizes expected regret and that regret is below an override budget; otherwise, it executes the host mask. We evaluate TRACE on an AI2-THOR 5 port of the full ALFRED validation splits, using expert high-level plans, teleport navigation, and hosts adapted from FILM and CAPEAM. With OpenAI Decisions, TRACE raises seen-split closed-loop success from 0.500 to 0.649 for the FILM-adapted host and from 0.528 to 0.659 for the CAPEAM-adapted host. The gains are concentrated in first-put grounding, such as selecting a sink basin rather than its rim. A controlled study shows that changing recovery costs changes override decisions even when beliefs are held fixed; a fixed probability threshold cannot reproduce this behavior.**

***Index Terms*—Cognitive robotics, decision making, decision theory, natural language processing, object recognition**

I. Introduction

In ALFRED [1], a language instruction names an object class, but every interaction action requires a pixelwise mask of one physical instance. Detecting the class is therefore insufficient: the selected mask must also satisfy the current subgoal and action history. For a sink-placement subgoal, the first `Put` must target the basin rather than the rim; the second pickup in Pick Two & Place must avoid the apple already moved. These examples illustrate two requirements for mask selection: grounding the instruction in the correct target and selecting the same or a different instance relative to earlier interactions.

Modular hosts FILM [2] and CAPEAM [3] deterministically select the closest mapped instance and highest-scoring box, respectively. Confidence alone cannot determine when to override these defaults: choosing an incorrect nearby drawer adds two interactions, whereas choosing an incorrect refrigerator in another room requires a round trip. The same peak belief 0.6 can therefore warrant different decisions that a fixed probability threshold cannot distinguish. The decision requires expected extra cost relative to the best alternative under each identity hypothesis [4], measured in interaction and navigation cost units.

We propose TRACE (Task-Regret-Aware Consequence Evaluation for Instance Association), a drop-in module for interaction-mask selection. TRACE derives a task relation from the history of successful actions, obtains candidate beliefs from a decision model, and uses a symbolic critic to estimate each candidate's extra interaction and navigation cost under each referent hypothesis. It weights these costs by the beliefs to compute expected regret. TRACE selects the unique minimizer only when its regret is below an override budget; otherwise, it immediately executes the host mask. The host retains navigation, semantic mapping, language parsing, and high-level planning.

We evaluate TRACE on an AI2-THOR 5 port of the full ALFRED Oct-21 validation splits (820 seen and 821 unseen episodes across all seven task types), using expert high-level plans and teleport navigation. The hosts are adapted from FILM and CAPEAM; the decision models Jev and OpenAI Decisions receive identical questions and images. The closed-loop gains are concentrated in first-put grounding, where selecting the wrong receptacle part can terminate an episode.

We make two contributions:

- **Formulation:** we formulate instance selection in terms of expected regret computed from candidate beliefs and candidate-specific extra costs.
- **Adapter:** we develop a modular adapter that uses an override budget to determine when to change the host's default selection.

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II. Related Work

A. Embodied Instruction Following

ALFRED [1] benchmarks natural-language instructions that require navigation and object interaction in AI2-THOR [5]. Each interaction action requires a pixelwise mask of one object, and an episode can include state changes through heating, cleaning, and slicing. The agent must therefore remember which drawer it opened, which potato it heated, and which instance it already picked up.

FILM [2] parses the instruction into a subtask list and builds a semantic map from depth and Mask R-CNN [6] instance segmentation. If the map already contains an object of the target class, FILM navigates to the closest observed instance, and a deterministic low-level policy performs the interaction. CAPEAM [3] extends a similar modular architecture with context-aware planning and environment-aware memory. It reuses the latest mask when recognition fails, excludes an already relocated instance from later targets, and caches the location and mask of an object whose state has changed. Its interaction executor still commits to a single Mask R-CNN box. Both agents map an object class to an interaction mask using deterministic rules; neither rule estimates the additional task cost of selecting an incorrect instance.

B. Instance Association and Decision-Making under Uncertainty

Instance association matches a previously manipulated object to a mask in the current view [7], for example, to identify and exclude the first apple when selecting the second in Pick Two & Place. First-put instruction grounding poses a distinct problem: the selected mask must satisfy the instruction, such as targeting a sink basin rather than its rim, without a remembered instance to match. FILM and CAPEAM handle both with geometry or stored interaction records, and neither returns an identity probability over the current candidate masks [2], [3].

Appearance embeddings such as DINOv2 [8] can score identity by crop similarity, and recent decision models return a typed distribution over a fixed option set from text and images [9], [10]. Neither a similarity score nor such a distribution alone specifies whether to override the host, because an incorrect selection can incur different costs across scenes even at the same probability. Classical decision theory supplies the missing quantity: the regret of a choice relative to the best alternative under each hypothesis [4]. TRACE weights the extra costs by the reported beliefs and compares the resulting expected regret with an override budget.

III. Methodology

TRACE is inserted at the host call that maps an object class to an interaction mask (Fig. 1); the host retains its mapping, navigation, language parsing, and planning components. The call has three stages: deriving a task relation from the history of successful actions, estimating a multimodal belief over the candidate masks, and applying a cost-based gate to select a candidate or execute the host mask.

TABLE I
Rules for determining the task relation r_t.

Rule	History condition	Next action	Relation
R0	No remembered instance of the target class	Pickup / Put	Ground
R1	Opened receptacle R	Put into / close R	Same
R2	Put into R	Close / toggle / reopen R	Same
R3	Toggled X on	Toggle X off	Same
R4	Put object into microwave / fridge / sink	Pickup of the same class	Same
R5	Put object elsewhere, or first Pick Two pickup	Pickup of the same class	Diff
—	No matching rule	—	None

A. Task Formulation and Identity Relations

At each interaction, TRACE receives the action a, the target class, a candidate list of N masks, the current RGB frame, a history of successful interactions, and a symbolic state s. It returns Select i with $i \in \{0, \ldots, N-1\}$ or Return; in the latter case, the host immediately executes the mask at its default index i_{host}. The current adapter invokes TRACE only for Pickup and Put when $N \geq 2$. Toggle and Slice execute the host mask, and Heat, Cool, and Clean use nearest-appliance scripts without TRACE. When $N{=}1$, the host executes the sole candidate without calling a decision model.

The adapter constructs the candidate list from simulator metadata: objects of the target class for Pickup and a receptacle pool for Put, without visibility filtering. It retains at most eight candidates, prioritizing the class named in the instruction. The FILM-adapted host selects the listed candidate closest to the agent [2]. The CAPEAM-adapted host scores each candidate box by the maximum product of detector confidence and intersection over union (IoU) among 119-class Mask R-CNN predictions with $\text{IoU} \geq 0.2$ [3], [6]; if none qualifies, it falls back to agent distance. Both are scorers adapted from the published agents, not their controllers, and TRACE shares each host's truncated list.

The task relation r_t is Same, Diff, Ground, or None. TRACE determines r_t from successful interaction history, task context, and the next action (Table I); simulator instance identifiers enter neither the relation rules nor the prompt. Same requires the next mask to correspond to the remembered instance, such as the drawer that the agent opened; Diff requires a different instance of the same class, as on the second pickup in Pick Two & Place. Ground specifies instruction grounding: memory contains no instance of the target class, and the selected mask must satisfy the instruction and subgoal, such as a sink basin rather than its rim. None leaves the host mask in place. R0 applies when memory contains no successful action on the target class; otherwise, the rules are evaluated in a fixed first-match order. In the current port, only R0, the Put branch of R1, and R5 reach the gate.

Memory records every action accepted by the simulator, including task-incorrect actions, without correction using

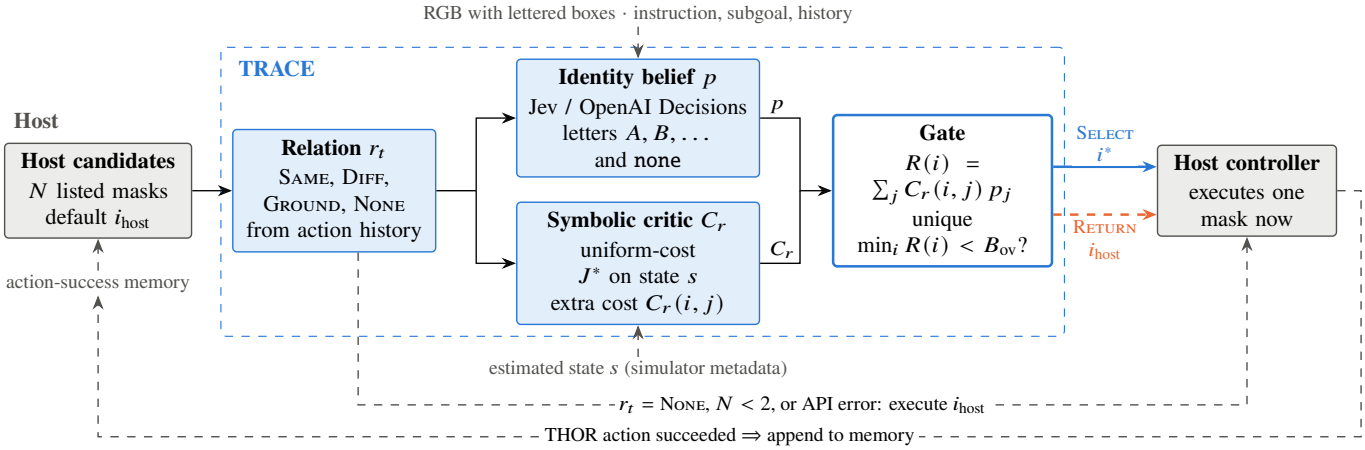
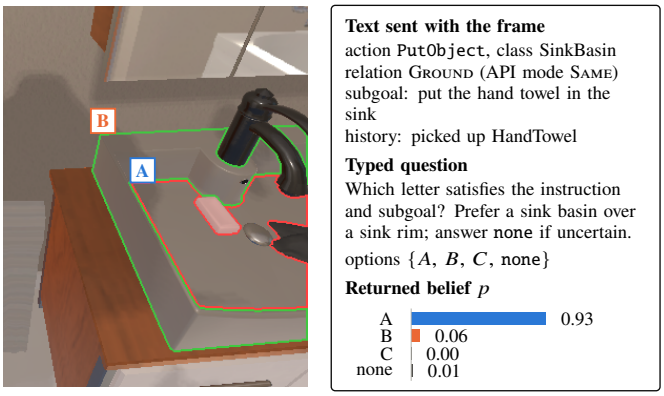


Fig. 1. Overview of TRACE for interaction-mask selection.



sink crop of the 300×300 frame; letters enlarged for print

Fig. 2. Example Jev input and output: lettered image (left), prompt and returned belief p (right).

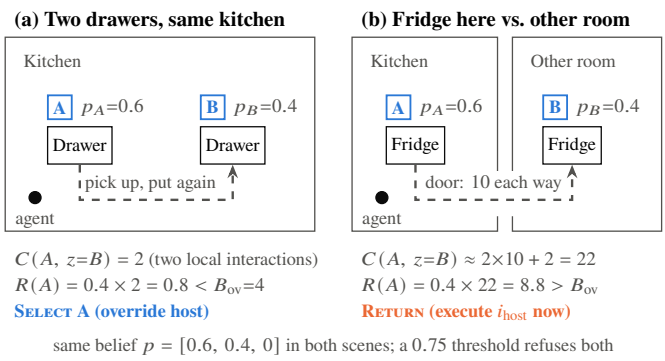


Fig. 3. Extra costs in two scenes with the same identity belief.

ground-truth identifiers. A pickup that the simulator accepts can still be the wrong apple, and that apple then becomes the referent for a later SAME or DIFF; action success is therefore a noisy proxy for task correctness.

B. Multimodal Instance Association

TRACE calls a decision model only for SAME, DIFF, or GROUND with $N \geq 2$. It submits one typed choice question whose options are the lettered candidates $A, B, \ldots$ in list order and `none` (Fig. 2). The visual input is the current RGB frame with a lettered box for each candidate; the text input contains the instruction, subgoal, a short history of successful actions, and relation mode. For SAME and DIFF, the question asks which candidate corresponds to the object involved in the earlier interaction, with `none` for an absent referent. For GROUND, it asks which candidate satisfies the instruction and subgoal, explicitly prefers a basin over a rim, and permits `none` when the model is uncertain. No remembered crop, expert reference image, or simulator identifier is sent.

Jev [9] and OpenAI Decisions [10] receive the same questions and images; each run uses one model at its default temperature without resampling. We ℓ_1-normalize the returned option probabilities to $p \in \mathbb{R}^{N+1}$, where p_j for $j < N$ is the probability that candidate j is the referent and p_N is the `none` mass. A refusal produces a uniform p and explicitly triggers RETURN; an API error triggers execution of i_{host} and is not counted as a TRACE mistake. We treat p as a reported belief, not a calibrated frequency; the critic treats p_N as a missing-referent hypothesis and adds a virtual object rather than switching off.

C. Consequence-Aware Commitment

The belief p alone cannot decide whether to override the host, because the cost of a wrong mask depends on the scene (Fig. 3). In the symbolic model, an incorrect mask selection can require additional pickups, room transitions, and puts. We define the extra cost relative to the cost of the best alternative under the same referent hypothesis. Selecting an incorrect open drawer in the same kitchen requires an additional pickup and put, for an extra cost of 2. Selecting a refrigerator in another room can require two room transitions at cost 10 each and two additional interactions, for an extra cost of 22.

The critic evaluates candidates using a symbolic state s constructed from simulator metadata, including object poses,

types, open/closed flags, and the held object. It groups locations into `here`, within 2.5 m of the agent, and `far`; a virtual object is placed in `elsewhere` under the missing-referent hypothesis. It reads neither expert identifiers nor task-success labels, and C_r estimates extra interaction and navigation cost, not the full ALFRED task cost. Let $z{=}j$ denote the hypothesis that candidate j is the referent and $z{=}N$ the hypothesis that no listed candidate is the referent. The adapter determines the critic objective from the action and r_t:

$$C_r = \begin{cases} C_{\text{put}}, & a = \texttt{PutObject}, \\ C_{\text{not-ref}}, & a = \texttt{PickupObject} \text{ and } r_t = \textsc{Diff}, \\ C_{\text{ref}}, & \text{otherwise.} \end{cases} \tag{1}$$

A SAME or GROUND pickup aims to hold the hypothesized referent; a DIFF pickup aims to hold a different instance of the same class (the `pickup_not_ref` objective, $C_{\text{not-ref}}$ in (1)). Thus, a belief concentrated on the first apple assigns a high expected regret to selecting that apple for a DIFF pickup.

Let $T(s, a_i)$ be the successor state after acting on mask i, $\text{cost}(a_i)$ the immediate action cost, and J^* the remaining cost to the critic goal. Committing to i when $z{=}j$ costs

$$\text{total}_r(i, j) = \text{cost}(a_i) + J^*\big(T(s, a_i) \mid z{=}j\big), \tag{2}$$

$$\text{base}_r(j) = \min\Big(J^*(s \mid z{=}j),\ \min_k \text{total}_r(k, j)\Big), \tag{3}$$

$$C_r(i, j) = \text{total}_r(i, j) - \text{base}_r(j), \tag{4}$$

where subtracting $\text{base}_r(j)$ removes the cost that even the best option under j would pay, so $C_r(i, j) \geq 0$. The matrix is $N \times (N{+}1)$ with the last column for $z{=}N$. We estimate J^* using uniform-cost search with interaction cost 1, room-transition cost 10, a horizon of 24, at most 5000 node expansions, and a cost of 40 for unreachable goals; these values are design constants. A sink rim is unreachable when the list also holds a basin, because a Put on the rim cannot succeed in THOR.

The expected regret of candidate i is the belief-weighted sum of its extra costs across referent hypotheses,

$$R(i) = \sum_{j=0}^{N} C_r(i, j)\, p_j, \tag{5}$$

and the gate selects a candidate only when the minimizer of expected regret is unique and its regret is strictly below the override budget B_{ov}:

$$i^* = \arg\min_i R(i), \qquad \begin{cases} \textsc{Select}\ i^*, & \text{unique and } R(i^*){<}B_{\text{ov}}, \\ \textsc{Return}, & \text{otherwise.} \end{cases} \tag{6}$$

RETURN triggers neither re-observation nor replanning; the host executes i_{host} at the current step. We set $B_{\text{ov}}{=}4$, equivalent to two local recoveries of cost 2 each; ties retain i_{host}. Fig. 4 compares the two costs at the same belief, $p_i{=}0.6$, with the remaining mass assigned to one incorrect candidate. An extra cost of 2 gives $R(i){=}0.8$, so TRACE selects the unique minimizer; an extra cost of 22 gives $R(i){=}8.8$, so TRACE returns control to the host.

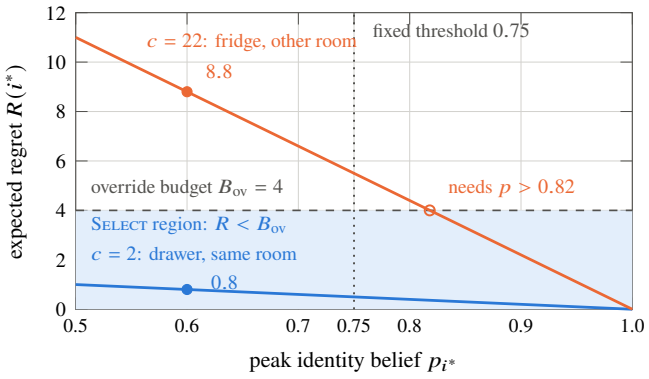


Fig. 4. Expected regret versus peak identity belief for two extra costs.

IV. EXPERIMENTS

A. Setup

We inserted TRACE into the FILM-adapted and CAPEAM-adapted scorers and ran the full ALFRED Oct-21 validation splits (820 seen and 821 unseen episodes across all seven task types) in AI2-THOR 5. Table II reports the closed-loop results, and Fig. 5 illustrates two interaction decisions.

The agent followed the expert high-level plan and teleported near the selected object before executing the interaction. The metadata predicate `goal_satisfied()` determines success, so the reported success rate (SR) measures performance under a fixed expert plan and teleport navigation. This setup differs from the official ALFRED protocol; its SR is not directly comparable with published ALFRED test-set results. Four selectors shared each host's candidate list: the host baseline, a ground-truth-identifier (GT-ID) selector that follows the expert pose when the corresponding mask is listed, TRACE + Jev, and TRACE + OpenAI Decisions. A DIFF pickup was evaluated with the `pickup_not_ref` objective; Toggle and Slice executed the host mask.

We report SR and, for calls with $N \geq 2$, the number of SELECT decisions (Sel.) and commitment regret (CR). Selection counts include decisions that retain the host mask. CR subtracts the oracle's scored cost from the scored cost of the selected decision, with RETURN assigned $B_{\text{ov}}{=}4$ before subtraction. Thus, CR evaluates decisions under this scoring rule and does not measure realized closed-loop extra cost. In a controlled study, we fixed $p{=}[0.6, 0.4, 0]$ and averaged results over a low-cost scene (wrong-mask extra cost 2) and a high-cost scene (extra cost 20). Detector scores of $[0.4, 0.9]$ made the host select the lower-probability candidate. For Table IV, fixed scoring assigns B_{ov} to RETURN, whereas host scoring uses the extra cost of the executed host mask. We also used three synthetic drawings to compare the modes of the distributions returned by Jev and OpenAI Decisions.

B. Results

TRACE improves closed-loop SR over both host baselines on the seen and unseen splits (Table II). TRACE + OpenAI Decisions achieves seen-split SRs of 0.649 and 0.659 for

TABLE II
CLOSED-LOOP RESULTS IN THE AI2-THOR 5 PORT. BOLD MARKS THE HIGHEST SR; DASHES INDICATE INAPPLICABLE COUNTS.

	FILM-adapted host				CAPEAM-adapted host			
	Seen (820)		Unseen (821)		Seen (820)		Unseen (821)	
Method	SR ↑ (net ep.)	Sel.	SR ↑ (net ep.)	Sel.	SR ↑ (net ep.)	Sel.	SR ↑ (net ep.)	Sel.
Host (baseline)	0.500	—	0.614	—	0.528	—	0.629	—
GT-ID selector	0.509	—	0.621	—	0.512	—	0.621	—
TRACE + Jev	0.615 (+94)	434	**0.636** (+18)	301	0.615 (+71)	431	**0.649** (+17)	294
TRACE + OpenAI	**0.649** (+122)	475	0.633 (+16)	318	**0.659** (+107)	477	0.642 (+11)	314

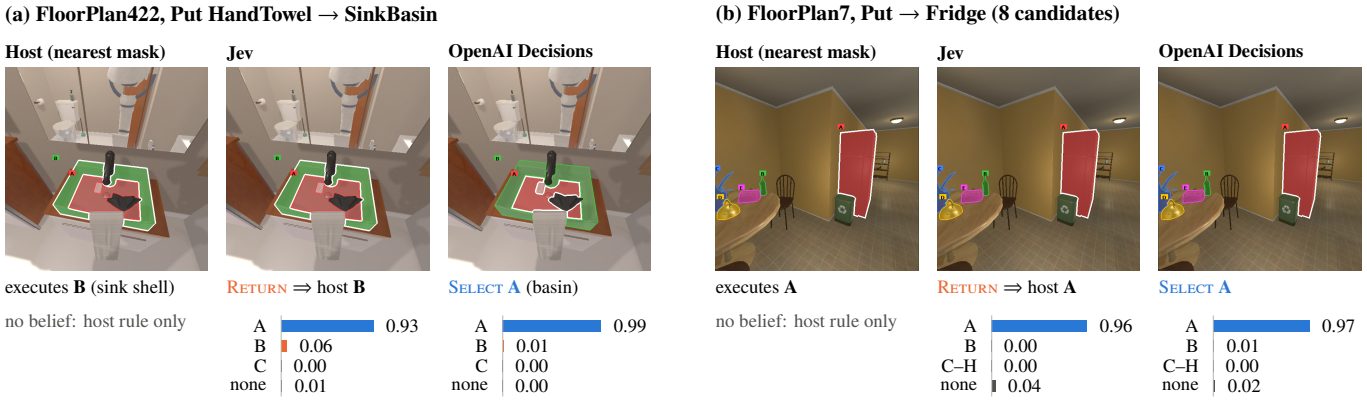


Fig. 5. Two closed-loop Put decisions for the host, Jev, and OpenAI Decisions: (a) HandTowel into SinkBasin in FloorPlan422; (b) Put into Fridge in FloorPlan7.

the FILM-adapted and CAPEAM-adapted hosts, respectively, compared with host SRs of 0.500 and 0.528. These improvements correspond to net gains of 122 and 107 successful episodes, respectively. On the unseen split, TRACE + Jev achieves the highest SRs, 0.636 (+18 episodes) and 0.649 (+17 episodes), for the FILM-adapted and CAPEAM-adapted hosts, respectively. TRACE + OpenAI Decisions achieves 0.633 (+16 episodes) and 0.642 (+11 episodes), respectively. TRACE + Jev achieves an SR of 0.615 on both seen splits, with net gains of 94 and 71 successful episodes for the FILM-adapted and CAPEAM-adapted hosts, respectively. The positive net gains in Table II indicate that episodes in which TRACE succeeds and the host fails outnumber those with the reverse outcome. The GT-ID selector follows expert poses, including those associated with sink rims that cannot receive objects in this port. Its SR is therefore not an upper bound on task success. Pickup success is 0.989–0.994 for every selector; the remaining failures involve Put, Heat, or Pick Two.

The gains are concentrated in first-put grounding, where the candidate list contains receptacles of the same class or parts of the same fixture, and an incorrect Put can terminate the episode (Table III). Look is 1.000 for every selector (omitted from Table III). Heat remains difficult: retrieval uses a host script, and the list often lacks the named microwave. With `pickup_not_ref` for DIFF pickups, TRACE + OpenAI Decisions raises seen-split Pick Two SR from 0.048 to 0.121 (FILM) and from 0.097 to 0.137 (CAPEAM).

TABLE III
SEEN-SPLIT SR BY TASK TYPE. BOLD MARKS THE HIGHEST SR IN EACH HOST GROUP.

	FILM-adapted (seen)			CAPEAM-adapted (seen)		
Task	Host	Jev	OpenAI	Host	Jev	OpenAI
Pick and Place	0.507	0.775	**0.880**	0.570	0.761	**0.880**
Movable recept.	0.435	0.539	**0.609**	0.391	0.530	**0.609**
Pick Two	0.048	0.097	**0.121**	0.097	0.097	**0.137**
Clean	0.705	**0.866**	**0.866**	0.839	**0.920**	**0.920**
Heat	0.065	**0.140**	0.121	0.047	**0.121**	0.112
Cool	0.810	0.905	**0.937**	0.810	0.897	**0.944**

The gate and probability thresholds make different decisions under the same belief (Table IV): a low threshold selects a candidate in both scenes, a high threshold returns control in both scenes, and TRACE selects a candidate only in the low-cost scene, where $R(i) < B_{\text{ov}}$. This comparison establishes different decision patterns, but does not establish lower realized extra cost for TRACE when RETURN executes the host mask. Two closed-loop examples illustrate the distinction between a gate decision and the executed mask (Fig. 5). For the HandTowel Put in FloorPlan422, OpenAI Decisions assigns approximately 0.99 to the basin and selects it, whereas Jev assigns approximately 0.93 and returns control to the host, which selects the rim. For the Put in FloorPlan7, Jev returns control to the host and OpenAI Decisions selects candidate A; both execute the same host mask.

TABLE IV
COMPARISON OF DECISION RULES. BOLD MARKS THE LOWEST MEAN COST.

Rule	Low cost	High cost	Mean (fixed) ↓	Mean (host) ↓
Host score	SELECT 1	SELECT 1	6.6	6.6
Threshold τ=0.55	SELECT 0	SELECT 0	4.4	**4.4**
Threshold τ=0.75	RETURN	RETURN	4.0	6.6
TRACE	SELECT 0	RETURN	**2.4**	6.4

TABLE V
PUT COMMITMENT REGRET (CR ↓). BOLD MARKS THE LOWEST CR.

	FILM-adapted		CAPEAM-adapted	
Method	Seen	Unseen	Seen	Unseen
Host (baseline)	2.74	**0.44**	3.41	1.92
Always-return	3.74	3.67	3.73	3.66
TRACE + Jev	**1.64**	1.38	**1.62**	**1.47**
TRACE + OpenAI	3.00	1.66	2.99	1.74

Both decision models identify the intended option in all three synthetic drawings (accuracy 1.00, top-1 agreement 1.00). Their reported beliefs differ most in the absent-referent example (L_1=0.64): Jev assigns 0.88 to `none`, whereas OpenAI Decisions assigns 0.56. TRACE makes the same decision with both models on each drawing. Jev has a lower Put CR than OpenAI Decisions in all four host–split settings and the lowest CR among all evaluated selectors in three (Table V). OpenAI Decisions makes more SELECT decisions and has a higher Put CR in all four settings, while achieving higher SR on the seen splits.

V. CONCLUSION

We presented TRACE, a module that selects interaction masks using candidate beliefs, an extra-cost matrix C_r, and an override budget. TRACE selected a candidate when the minimizer of expected regret was unique and its value was below the budget; otherwise, it executed the host mask. On an AI2-THOR 5 port of the full Oct-21 validation splits with expert plans and teleport navigation, TRACE + OpenAI Decisions improved seen-split SR from 0.500 to 0.649 for the FILM-adapted host and from 0.528 to 0.659 for the CAPEAM-adapted host, with gains concentrated in first-put grounding. The controlled study showed that cost-dependent decisions changed under a fixed belief and could not be reproduced by a fixed probability threshold. The evaluation is limited by the host candidate list, action-success memory, and the quality of p and C_r; improving these inputs is a direction for future work.

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